



ROBUST TRADING STRATEGY FOR BATTERY ENERGY STORAGE SYSTEM IN THE CZECH ELECTRICITY MARKET

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Instructions

This thesis aims to design a resilient operation strategy for a Battery Energy Storage System (BESS) participating in the Czech electricity market. The objective is to navigate the volatility of local prices with robust revenue generation strategy.

- 1) Acquire and preprocess historical market data from the OTE (Czech electricity market operator).
- 2) Design and implement two distinct decision-making frameworks: a baseline model that predicts prices for a given time period and then optimizes the charging and discharging of the BESS (i.e., a predict-then-optimize approach), and a robust alternative explicitly built to mitigate forecast errors and uncertainty (e.g., stochastic optimization).
- 3) Conduct a comparative simulation to evaluate the financial performance and operational reliability of both approaches, specifically analyzing how the robust strategy performs during periods of unexpected market fluctuations compared to the predict-then-optimize approach. Also, compare the results to:
 - (a) A simple strategy of discharging during well-known morning and evening peak hours and charging before those peaks.
 - (b) The theoretical maximal profit in the situation where the prices are known and only optimization needs to be performed.

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Declaration

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In Prague on May 15, 2026

Abstract

Electricity is one of the key resources of modern society, and its efficient, reliable, and sustainable production is becoming increasingly important. This creates a need for technologies that can store energy and help balance fluctuations in the power grid. One such technology is the Battery Energy Storage System (BESS), which can store electricity during periods of high generation or low prices and later supply it back to the grid when demand or prices are higher. Besides supporting the power grid, BESS operation also creates an opportunity for energy arbitrage in the electricity market. This thesis focuses on creating a profitable battery charging and discharging schedule that can be used for trading on the Czech day-ahead electricity market. Two main approaches are used to create this schedule. The first is a predict-then-optimize approach, which directly uses point price forecasts. The second is a robust alternative, which accounts for price uncertainty and aims to reduce potential losses. The implemented methods are then compared with simpler reference charging schedules and with the best possible schedule.

Keywords Battery Energy Storage System, Czech electricity market, day-ahead market, electricity price forecasting, energy arbitrage, BESS scheduling

Abstrakt

Elektřina je jedním z klíčových zdrojů moderní společnosti a její efektivní, spolehlivá a udržitelná výroba se stává stále důležitější. To vytváří potřebu vývoje technologií, které dokážou ukládat energii a tak pomáhat vyrovnávat výkyvy v elektrické síti. Jednou z takových technologií je Battery Energy Storage System (BESS), který může ukládat elektřinu v obdobích vysoké výroby nebo nízkých cen a následně ji dodávat zpět do sítě v době, kdy je poptávka nebo cena vyšší.

Kromě podpory elektrické sítě vytváří provoz BESS také příležitost pro energetickou arbitráž na trhu s elektřinou. Tato práce se zaměřuje na vytvoření ziskového plánu nabíjení a vybíjení baterie tak aby se dal využít pro obchodování na českém denním trhu s elektřinou. Pro vytvoření tohoto plánu jsou použity dva hlavní přístupy: První je predict-then-optimize, který přímo využívá bodové predikce cen. A alternativa v podobě robustního přístupu, která zohledňuje nejistotu cen a snaží se minimalizovat ztráty. Implementované metody jsou následně porovnány s jednoduššími referenčními plány nabíjení a také s nejlepším možným plánem.

Klíčová slova Battery Energy Storage System, český trh s elektřinou, denní trh s elektřinou, predikce cen elektřiny, energetická arbitráž, plánování provozu BESS

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List of abbreviations

BESS	Battery Energy Storage System
SOC	State of Charge
SDAC	Single Day-Ahead Coupling
OTE	Czech electricity and gas market operator
NEMO	Nominated Electricity Market Operator
MILP	Mixed-Integer Linear Programming
CVaR	Conditional Value-at-Risk
EnbPI	Ensemble Batch Prediction Intervals
MAPIE	Model Agnostic Prediction Interval Estimator
HGBR	Histogram-Based Gradient Boosting Regressor
RMSE	Root Mean Squared Error
MAE	Mean Absolute Error
wMAPE	Weighted Mean Absolute Percentage Error

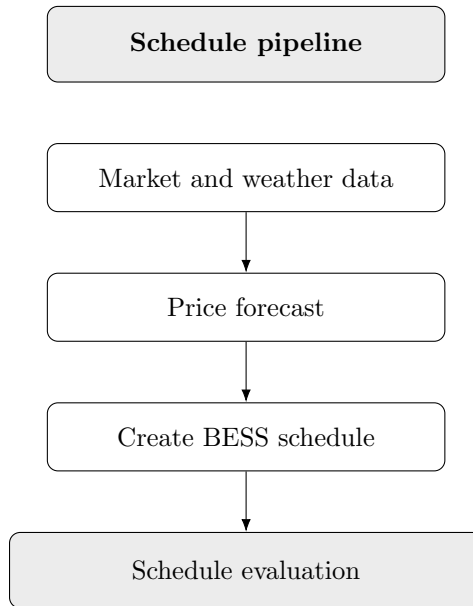
Introduction

This thesis focuses on the operation of Battery Energy Storage System (BESS) in the Czech electricity market, with emphasis on day-ahead price uncertainty, revenue optimization, and operational resilience. The work compares baseline predict-then-optimize strategy with a more uncertainty-aware optimization framework and evaluates their performance under realistic market conditions.

1.1 Motivation and Research objective

The growing share of renewable energy sources increases short-term variability in electricity systems and contributes to stronger price fluctuations in electricity markets. At the same time, BESS are becoming an important flexibility resource, since they can shift energy in time and respond quickly to market signals. The motivation of this thesis is to examine how a BESS can be operated profitably in the Czech day-ahead electricity market and whether uncertainty-aware scheduling can improve the resulting performance.

The main objective of this thesis is to design and evaluate a schedule creation pipeline for the operation of a BESS participating in the Czech day-ahead electricity market. The general structure of the framework is shown in Fig. 1.1. This simplified pipeline serves as a starting point for the thesis and is described in more detail in Chapter 4.



■ **Figure 1.1** Simplified schedule creation pipeline.

This main objective can be divided into several smaller objectives that guide the structure of the thesis:

- Acquire and preprocess historical day-ahead market data relevant to the Czech electricity market.
- Develop forecasting models for Czech day-ahead electricity prices.
- Implement a baseline forecast-driven scheduling approach.
- Implement an uncertainty-aware scheduling approach.
- Compare scheduling approaches on historical market data.
- Benchmark the obtained results against simple and idealized reference strategies.

Theoretical background

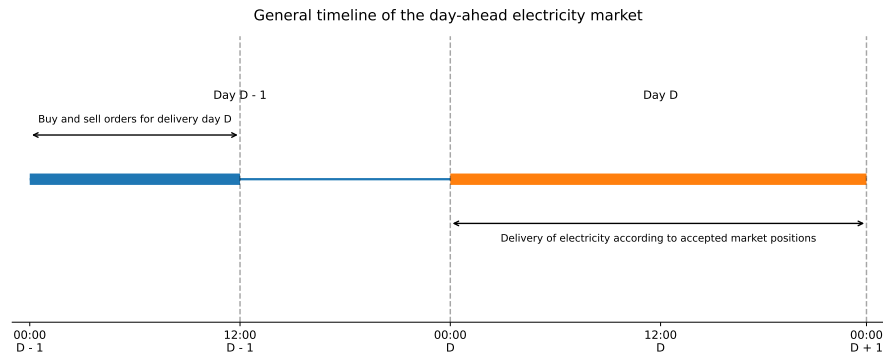
This chapter introduces the theoretical concepts needed for this thesis. It covers the structure of the day-ahead electricity market, the technical and economic characteristics of battery energy storage systems, forecasting methods for electricity prices, uncertainty quantification, and optimization approaches suitable for decision-making under conditions of uncertainty.

2.1 The European Electricity Market

The European electricity market is organized across several time horizons. The main market layers include the forward market, the day-ahead market, the intraday market, and the balancing mechanism. Market participants can act either as sellers or buyers of electricity. Sellers supply electricity to the power system, for example from conventional power plants, wind farms, or solar parks. Buyers purchase electricity either for direct consumption or for resale, and may include industrial consumers, electricity suppliers, traders, or BESS operators. In the context of this thesis and the optimization of the BESS schedule, the focus is on the day-ahead market.

The day-ahead market

The day-ahead market plays a central role, as it serves as the main reference market for electricity delivery the following day. Market participants can submit bids to buy or sell a specified amount of electricity for individual delivery periods, and these bids are matched through a market-clearing process. For the market on day D, this clearing process takes place at 12:00 on day D - 1. The basic timing of this process is illustrated in Fig. 2.1. This is crucial for the present thesis because all the input required to create an optimal BESS schedule must be prepared in advance so that the corresponding orders can be submitted before the day-ahead market closes.



■ **Figure 2.1** Timeline of the day-ahead electricity market.

Czech Market

The Czech Republic has been part of the 4M Market Coupling since 2014, together with Hungary, Slovakia, and Romania. In 2021, this structure was connected with the rest of Europe, and the Single Day-Ahead Coupling (SDAC) was created, which now connects almost all European countries. A general overview is shown in Figure 2.2. This solution increases trading efficiency through greater competition, higher liquidity, and better sharing of energy across Europe. In 1 October 2025 SDAC launched 15-minute market [1]. This allows market participants to offer or buy electricity every 15-minute period of the following day.

This change allows the market to reflect real-time conditions more accurately. It is particularly important for renewable energy sources, whose output can be variable and change rapidly, because shorter trading intervals allow more precise scheduling and dispatch. A finer time resolution also improves forecasting and balancing of supply and demand, which helps reduce imbalances and enhance overall grid reliability. In addition, it increases market efficiency by allowing participants to trade in a more granular way and respond more effectively to market signals.



■ **Figure 2.2** Geographical scope of the SDAC. Source: NEMO Committee [2].

OTE

OTE, a.s. is the Czech market operator that provides services in the energy sector to participants trading energy commodities. Its activities include, among others, the organization of trading on the day-ahead and intraday electricity markets, including continuous trading and intraday auctions. OTE also acts as the Nominated Electricity Market Operator (NEMO) for the Czech Republic and co-organizes the single European day-ahead and intraday electricity markets. In addition, it evaluates imbalances in the electricity and gas markets, ensures clearing and settlement for market participants, processes market data, and publishes market results [3].

For the purposes of this thesis, OTE is important mainly because it operates the organized Czech short-term electricity market and publishes day-ahead market results that are used as the main source of historical price data. The day-ahead market is an auction-based market in which market participants submit purchase and sale orders for delivery on the following day. According to the current OTE market parameters [4], day-ahead trading is denominated in EUR, the minimum tradable volume is 0.1 MW, the maximum tradable volume is 2,999 MW, and the allowed price range is from -500 EUR/MWh to $4,000$ EUR/MWh.

OTE also operates short-term gas markets in the Czech Republic. This is relevant in the broader context of electricity price forecasting because natural gas prices can influence electricity prices in European power markets, especially when gas-fired power plants set the marginal generation cost.

2.2 Day-ahead electricity price characteristics

Day-ahead electricity prices are strongly affected by intraday and seasonal patterns. These patterns are mainly driven by changes in electricity demand, renewable generation, fuel prices, and the availability of generation capacity. In general, higher demand tends to increase prices because more expensive generation sources may be needed to satisfy consumption. In contrast, higher renewable generation can reduce prices, since wind and solar power have low marginal generation costs.

The renewable generation is generally negatively correlated with electricity prices. When solar and wind production exceed the flexibility of the electricity system, the probability of very low or negative prices increases [5]. This makes renewable generation forecasts, together with load forecasts, important explanatory variables for day-ahead electricity price forecasting.

The average intraday profiles of the day-ahead price, forecasted load, and forecasted renewable generation used in this thesis are shown in Fig. 2.3.

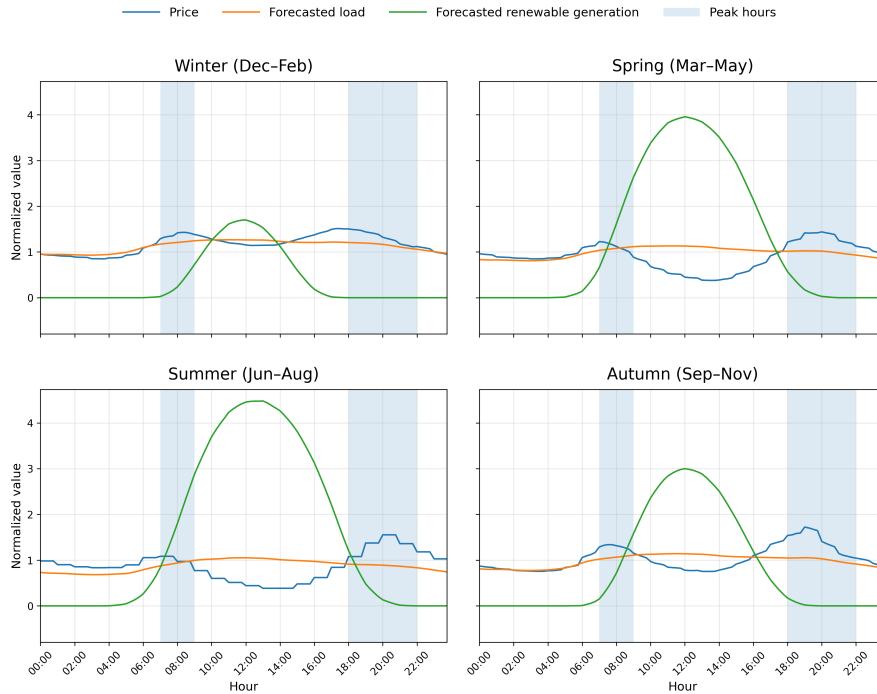
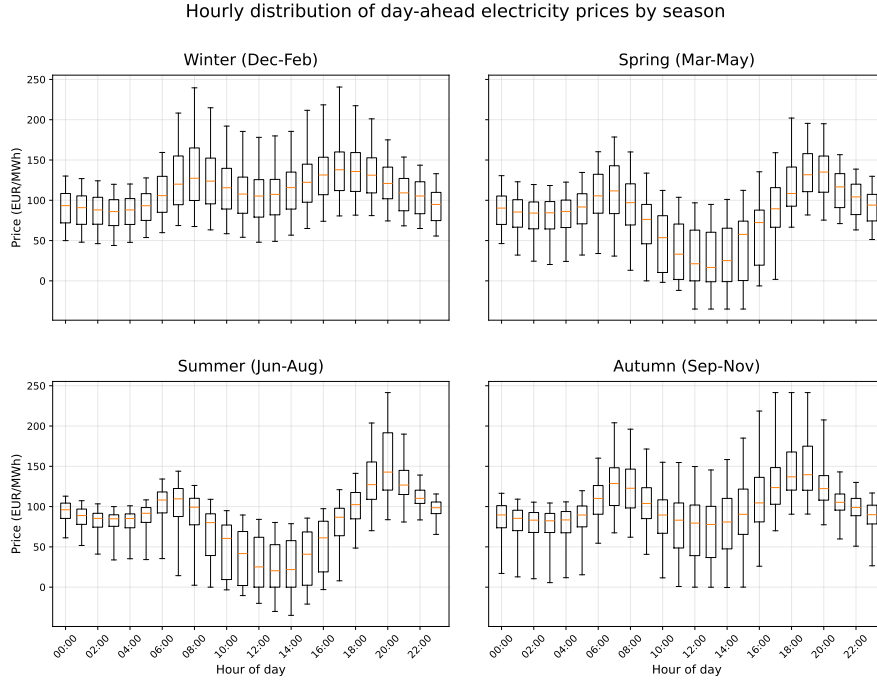


Figure 2.3 Average profiles of day-ahead price, forecasted load, and forecasted renewable generation.

Fig. 2.4 shows the hourly distribution of day-ahead electricity prices separately for individual seasons. The exact timing and strength of the price peaks change during the year, the pattern remains visible. In most seasons, prices tend to be high during before noon and evening hours. This supports the use of a simple peak-hour schedule (explained in 4.3.1) as a strategy, since it reflects the recurring daily pattern observed in the historical price data.



■ **Figure 2.4** Seasonal hourly distributions of day-ahead electricity prices.

2.3 Point forecasting of day-ahead electricity prices

The objective of the day-ahead price forecasting task is to predict the sequence of electricity prices for the following trading day. In the implemented framework, this corresponds to a 24-hour horizon divided into 15-minute intervals. In the first stage, the considered models are used for point prediction, with the aim of identifying the model that achieves the lowest RMSE on the available historical data. In the practical part of the thesis, the following models are evaluated: **Linear Regression**, **Random Forest**, **XGBoost**, **Histogram-Based Gradient Boosting Regressor (HGBR)**, and **CatBoost**. These models are also used as base estimators in the MAPIE framework, specifically with the Ensemble Batch Prediction Intervals (EnbPI) method, to construct prediction intervals.

As baseline methods, this thesis considers a linear model and a random forest model. These approaches were selected because the linear model serves as a simple and interpretable benchmark and the random forest represents a more flexible nonlinear ensemble method. Together, they provide a useful reference point for evaluating the performance of more advanced boosting-based models.

2.3.1 Linear model

The linear model used in this thesis is ordinary least squares regression. It assumes that the predicted electricity price can be expressed as a linear combination of the input features,

$$\hat{y}(x) = w_0 + w_1x_1 + \cdots + w_px_p,$$

where $x_1, \dots, x_p$ are the input variables, $w_1, \dots, w_p$ are the corresponding coefficients, and w_0 is the intercept. The model parameters are estimated by minimizing the residual sum of squares between the observed and predicted values. Due to its simplicity and interpretability, the linear model is used as a baseline for comparison with more complex machine learning methods. [6]

2.3.2 Random Forest

Random forest is an ensemble learning method based on a large number of decision trees. Each tree is trained on a bootstrap sample of the training data, and at each split only a random subset of features is considered. The final prediction is obtained by averaging the predictions of all trees. This randomization reduces the correlation between individual trees and improves the generalization ability of the ensemble [7]. Compared with the linear model, random forest can capture nonlinear relationships and interactions between features without requiring an explicit functional form.

2.3.3 Boosting models

Boosting is an ensemble learning approach in which weak learners are added sequentially so that each new model reduces the errors of the previous ensemble. In gradient boosting, this process can be interpreted as a stage-wise optimization procedure in function space, where the final model is constructed as an additive expansion of base learners, most commonly decision trees. A simplified form of the model can be written as

$$F_M(x) = \sum_{m=1}^M \gamma_m h_m(x),$$

where $h_m(x)$ is the m -th tree, γ_m is its weight, and M is the total number of boosting iterations. In each step, the new tree is fitted so that it improves the current model with respect to a chosen loss function. [8]

In this thesis, three boosting-based models are considered: HGBR, CatBoost, and XGBoost.

2.3.4 Histogram-Based Gradient Boosting Regressor

Histogram-Based Gradient Boosting Regressor (HGBR) is a gradient boosting implementation available in scikit-learn [9]. Its main idea is to first discretize continuous features into a finite number of bins and then search for tree splits over these histogram bins instead of over all original feature values. This significantly reduces the computational cost of training, especially on large tabular datasets. This characteristic was useful in this thesis, because when testing different forecasting approaches, especially in combination with EnbPI, it was advantageous to use a fast model in the initial stages of selecting the best-performing method. The scikit-learn implementation is explicitly described as being inspired by LightGBM [10], which introduced histogram-based learning as an efficient improvement over classical gradient boosting decision trees.

2.3.5 CatBoost

CatBoost is a gradient boosting method based on decision trees that differs mainly in its training procedure and treatment of categorical features. Its key idea is *ordered boosting*, which reduces target leakage and prediction shift by processing observations in a permutation-based order, so that each prediction is based only on information that would have been available beforehand. In addition, CatBoost is designed to handle categorical variables efficiently without requiring simple one-hot encoding. This is particularly useful in this thesis, since several input features, such as `is_daylight`, `is_holiday`, `dow`, `month`, `qod`, and `slot_week`, can naturally be interpreted as categorical or discrete variables.

2.3.6 XGBoost

XGBoost is a scalable gradient boosting system that extends the standard boosting framework with regularization and several computational improvements. Its objective function combines a training loss term with a regularization term on tree complexity, which helps control overfitting. In addition, XGBoost uses both first-order and second-order derivatives of the loss function when constructing trees, which makes the optimization more precise than simple gradient-based updates. [11]

2.4 Interval forecasting

Point forecasts alone are insufficient in applications where price uncertainty is relevant. A prediction interval provides a lower and an upper bound, denoted by $\hat{y}^L$ and $\hat{y}^U$, around the point forecast. Ideally, this interval should satisfy two competing requirements. On the one hand, the empirical coverage, defined in Section 4.4.1, should be close to the chosen confidence level. In this context, the confidence level represents the expected share of true prices that should fall inside the prediction interval. For example, for a 95% confidence level, approximately 95% of realized prices should lie between the lower and upper prediction bounds. On the other hand, the interval width should remain as small as possible, since overly wide intervals are less informative and less practical for further use.

Interval forecasting complements the point prediction with lower and upper bounds that quantify forecast uncertainty. This is the first step in scenario generation for the stochastic scheduling framework. In this thesis, two approaches are used for the creating prediction intervals.

2.4.1 Quantile-based prediction intervals

For the direct construction of prediction intervals, a quantile-based approach is used. Three separate models are trained for each forecasting method. The first model is used for point prediction and estimates the expected value of the target variable. The other two models estimate the lower and upper bounds of the prediction interval by learning selected conditional quantiles.

The point forecast is obtained from a model trained with a standard regression loss function, such as squared error. The lower and upper bounds are estimated by two additional models trained with quantile loss [12, 13]. The **Quantile** objective is defined for target values y_i , predictions $\hat{y}_i$, observation weights w_i , and quantile level $\alpha \in (0, 1)$ as [12]

$$L_\alpha = \frac{\sum_{i=1}^N w_i (\alpha - \mathbf{1}\{y_i \leq \hat{y}_i\})(y_i - \hat{y}_i)}{\sum_{i=1}^N w_i}.$$

This is the standard quantile regression loss, also known as the pinball loss. It penalizes overestimation and underestimation asymmetrically, with the degree of asymmetry controlled by the parameter α . Small values of α are suitable for estimating lower conditional quantiles, while values close to 1 are suitable for estimating upper conditional quantiles.

2.4.2 Ensemble Batch Prediction Intervals

EnbPI is a conformal method for constructing prediction intervals for time-series data. EnbPI is designed for settings where the order of observations

carries important information. This is suitable for this thesis because day-ahead electricity prices form a time series, where neighbouring observations are related. In the MAPIE implementation, EnbPI is described [14] as being similar to jackknife-after-bootstrap: leave-one-out predictors are approximated using a finite number of bootstrap models, while the prediction interval is constructed from the empirical distribution of the resulting residuals.

The main steps of the EnbPI procedure are [15]:

1. Given training data $\{(x_i, y_i)\}_{i=1}^n$, train B bootstrap predictors $\hat{\mu}_1, \dots, \hat{\mu}_B$.
2. For each training observation i , construct a leave-one-out aggregated prediction $\hat{\mu}_{-i}(x_i)$ by combining only those bootstrap models that were not trained on (x_i, y_i) .
3. Compute the leave-one-out residuals

$$R_i^{\text{LOO}} = y_i - \hat{\mu}_{-i}(x_i), \quad i = 1, \dots, n.$$

4. For a new input x_t , compute the aggregated prediction

$$\hat{\mu}_{\text{agg}}(x_t).$$

5. Using the most recent residual set $\{R_i^{\text{LOO}}\}$, choose $\gamma \in [0, \alpha]$ so that the prediction interval is as short as possible.
6. Construct the $(1 - \alpha)$ prediction interval as

$$[\hat{y}_t^L, \hat{y}_t^U] = [\hat{\mu}_{\text{agg}}(x_t) + \hat{q}_{n, \gamma}\{R_i^{\text{LOO}}\}, \hat{\mu}_{\text{agg}}(x_t) + \hat{q}_{n, 1-\alpha+\gamma}\{R_i^{\text{LOO}}\}].$$

7. Once the true value y_t becomes available, update the residual set and use the updated residuals for the next prediction interval.

2.5 Battery energy storage systems

BESS is an electrochemical device that stores electrical energy by charging from the grid or from a power plant and discharging it at a later time to provide electricity or other grid services when needed. In practice, there are many types of grid-scale battery system, which usually differ in the materials and technologies used. These systems can be built using lithium-ion, lead-acid, redox flow, and molten-salt batteries. Nowadays, lithium-ion technology dominates most practical applications. Similar to an ordinary battery, a BESS is mainly valued for its operational flexibility, fast response, and ability to charge and discharge quickly. These properties determine the range of applications for which a BESS can be used. The most common applications are [16]:

- **Arbitrage** – charging the battery when electricity prices are low and discharging when prices are high, or charging during periods of low demand when the grid has excess energy.
- **Firm or peaking capacity** – supporting the power system during periods of peak demand and improving the reliability and capacity value of renewable energy sources.
- **Operating reserves and ancillary services** – providing fast-response services such as frequency regulation, load-following, and ramping to help maintain the balance between generation and demand.
- **Transmission and distribution upgrade deferral** – reducing peak load on the grid and thereby postponing or avoiding costly network upgrades.
- **Black start** – supplying initial power needed to restart generators or parts of the grid after a system outage.

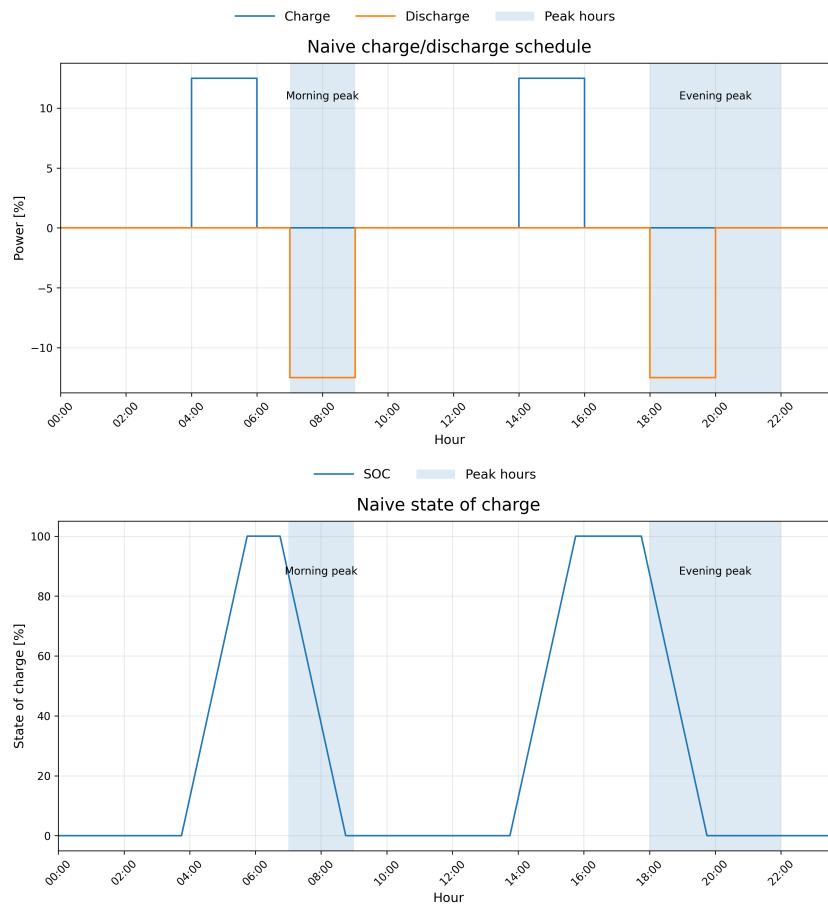
2.5.1 Technical Parameters of BESS

Because battery energy storage systems can differ widely in their design and technical characteristics, many parameters can be used to describe them. For the purposes of this thesis, the following parameters are sufficient [16]:

- **Energy capacity** ($\bar{E}$) is the maximum amount of energy that can be stored, usually expressed in kWh or MWh.
- **Charge limit** ($\bar{C}$) is the maximum amount of energy that can be charged into the BESS during one scheduling interval, usually expressed in kWh or MWh per interval.
- **Discharge limit** ($\bar{D}$) is the maximum amount of energy that can be discharged from the BESS during one scheduling interval, usually expressed in kWh or MWh per interval.
- **Cycle life** (N_{cyc} or L) is the number of charge–discharge cycles that the battery can undergo before its capacity degrades to an unacceptable level.
- **Self-discharge** (σ_{sd}) is the gradual loss of stored charge over time when the battery is not in active use, usually expressed as a percentage over a given period.
- **State of charge (SOC)** is the current level of charge relative to the full battery capacity, usually expressed as a percentage from 0 to 100%.
- **Round-trip efficiency** (η_{rt}) is the ratio of discharged energy to charged energy, including system losses, usually expressed as a percentage.

2.6 BESS schedule

A BESS schedule describes how the battery is operated over a given time horizon. In this thesis, the schedule specifies for each time interval whether the battery is charged–discharged. As a result, the amount of energy stored in the battery, is represented by the SOC. An example of a charge and discharge schedule, together with the corresponding state of charge, is shown in Fig. 2.5.



■ **Figure 2.5** Naive BESS operating schedule and the resulting state of charge.

For most of the BESS the schedule should not change from charging to discharging in every interval. Frequent and unnecessary charge–discharge movements increase battery throughput and accelerate degradation [17]. Since battery lifetime is limited by regular charging and discharging cycles, practical BESS operation usually has to balance short-term arbitrage profit against long-term degradation costs.

2.6.1 Optimization approaches for BESS scheduling

The creation of a BESS schedule is usually formulated as an optimization problem in which the charging and discharging decisions are selected so that a given technical or economic objective is achieved. In the literature, several optimization techniques have been used for this purpose, depending on the considered market setting, uncertainty treatment, and model complexity.

Mixed-integer linear programming

A common approach is mixed-integer linear programming (MILP). For example, Puech et al. [18] formulate day-ahead battery arbitrage as a MILP problem with operational and availability constraints and use it to determine the profit-maximizing charging and discharging schedule for the day-ahead market. This type of formulation is usable because it provides an exact optimal solution and can be solved using standard optimization solvers.

Dynamic programming

Another approach is dynamic programming. Li et al. optimize the schedule of a grid-connected photovoltaic-battery system by means of a dynamic programming algorithm, where the overall energy flow is controlled in real time in order to maximize the economic benefit of the system [19]. Dynamic programming is well suited to storage scheduling problems because the battery state of charge naturally creates a sequential decision structure across time.

Reinforcement learning

Reinforcement learning is another approach used for BESS schedule optimization. In this framework, battery operation is treated as a sequential decision-making problem. At each time step, the controller observes the current state of the system, and then selects an action such as charging, discharging, or remaining idle. After the action is applied, the controller receives a reward, usually defined according to the achieved operating profit, avoided cost, or another performance criterion [20].

The main idea is that the control policy is learned through repeated interaction with the environment. Instead of solving an explicit optimization problem for each scheduling horizon, the reinforcement learning agent gradually learns which actions are beneficial in different system states.

2.6.2 Linear programming in Pyomo

Pyomo represents optimization problems through algebraic modeling components. According to the official documentation [21], the standard modeling

workflow consists of creating a model and declaring its components, instantiating the model, applying a solver, and then interrogating the solver results. The same documentation states that the key components of a Pyomo model are sets, parameters, decision variables, objectives, and constraints. A simple linear problem shown in the Pyomo documentation is given by

$$\min 2x_1 + 3x_2 \quad \text{s.t.} \quad 3x_1 + 4x_2 \geq 1, \quad x_1, x_2 \geq 0,$$

where the objective is linear and the feasible region is defined by linear constraints in the decision variables.

A linear programming model therefore consists of three main parts. First, decision variables represent the unknown quantities to be chosen by the optimizer. Second, a linear objective function specifies what should be minimized or maximized. Third, linear equality and inequality constraints restrict the values that the decision variables may take.

In practical terms, linear programming works by searching for variable values that satisfy all model restrictions while producing the best value of the objective function. In a concrete Pyomo model, the algebraic expressions can be written directly through the `expr` argument, while in more general indexed models they are often generated through rule functions. After the model is defined, Pyomo passes it to a compatible external solver using `SolverFactory(...).solve(...)`. The solver returns a results object whose status and termination condition can then be checked to verify whether an optimal solution was found [22].

Data and preprocessing

This chapter describes the data used in the thesis. It presents the main data sources, explains the construction of the 15-minute forecasting dataset, and introduces the market, weather, calendar, and engineered features used as inputs for the forecasting models.

3.1 Data sources

The dataset was assembled from several sources. The target variable `15min_cena` (EUR/MWh) was obtained together with market and system variables from the ENTSO-E Transparency Platform using the `entsoe-py` library and its `EntsoePandasClient` interface [23]. These variables include `load_fc`, `solar_fc`, `wind_fc`, `generation_fc`, `consumption_fc`, and `de_price`.

Weather-related variables were downloaded from the Open-Meteo forecast and archive APIs [24, 25]. These include `temp_fc`, `cloud_fc`, `wind_speed_100m`, and `shortwave_radiation`. The solar-time features such as `min_since_sunrise`, `min_until_sunset`, and `is_daylight` were computed from the geographical location using the `pvlb` library [26].

The holiday indicator `is_holiday` was generated with the `holidays` Python library for Czech public holidays [27]. Finally, `gas_price` and `co2_price` were retrieved with the `yfinance` package from Yahoo Finance market data [28]. Additional artificial features, such as lagged values and rolling statistics, were then derived directly from the collected historical price series.

3.2 Data Used for Forecasting

The forecasting dataset is organized as a time series with a 15-minute resolution. It contains observations spanning the period from 1 January 2024 to 1 September 2025. Since the implemented forecasting and scheduling pipeline operates at a 15-minute resolution, each daily prediction horizon consists of

96 consecutive values. All historical hourly day-ahead price values before 1 September 2025 were transformed to 15-minute resolution by assigning the same hourly price to each of the four corresponding 15-minute intervals.

The first group consists of direct input features obtained from market data, generation forecasts, and meteorological forecasts. An important requirement is that all input variables must be available before the day-ahead market gate closure, so that they can realistically be used when creating the schedule for the following delivery day.

3.2.1 Market Features

These variables represent the fundamental factors that may influence electricity price formation. In particular, load and renewable generation forecasts describe the expected balance between demand and supply, while fuel and emission prices reflect the cost conditions of thermal generation. Weather variables are included mainly because they affect both electricity consumption and renewable production.

■ **Table 3.1** Market, system, and weather features used in the forecasting dataset

Feature	Description
load_fc	Forecasted electricity load.
solar_fc	Forecasted solar generation.
gas_price	Natural gas price.
co2_price	Price of CO ₂ emission allowances.
temp_fc	Forecasted temperature.
cloud_fc	Forecasted cloud cover.
wind_speed_100m	Forecasted wind speed at 100 meters.
shortwave_radiation	Forecasted shortwave solar radiation.
generation_fc	Forecasted total generation.
de_price	German electricity price.

Although `generation_fc` is formally a forecasted variable, in practice this series could not be obtained reliably in a stable day-ahead form. For this reason, `generation_fc` was shifted by one day and the value from day $D - 1$ was used as a proxy for day D . The same treatment was applied to `de_price`, so both variables enter the forecasting model as one-day lagged features.

3.2.2 Engineered price and calendar features

The second group of input variables consists of engineered features derived from the historical price series and from calendar or solar-cycle information.

These features are used to capture daily, weekly, and seasonal patterns in electricity prices. Price-based features are constructed only from historical values that would already be known before the forecasted delivery day.

■ **Table 3.2** Engineered price and calendar features used in the forecasting dataset

Feature	Description
<code>lag_k</code>	Historical electricity price observed k 15-minute intervals earlier, with $k \geq 96$. This ensures that only prices from previous days are used.
<code>y_mean_yday</code>	Mean electricity price of the previous day.
<code>y_std_yday</code>	Standard deviation of electricity prices from the previous day.
<code>y_spread_yday</code>	Difference between the maximum and minimum electricity price of the previous day.
<code>y_mean_7d</code>	Average electricity price over the previous seven days.
<code>quarter_of_day</code>	Index of the 15-minute interval within the day.
<code>hour</code>	Hour of the day.
<code>day_of_week</code>	Day of the week.
<code>month</code>	Month of the year.
<code>day_of_year</code>	Position of the day within the year.
<code>day_of_month</code>	Position of the day within the month.
<code>week_of_year</code>	Week number within the year.
<code>quarter_of_day_sin</code>	Sine transformation of the quarter-of-day index.
<code>quarter_of_day_cos</code>	Cosine transformation of the quarter-of-day index.
<code>qod_of_week</code>	15-minute interval index within the week.
<code>min_since_sunrise</code>	Number of minutes since sunrise.
<code>min_until_sunset</code>	Number of minutes until sunset.
<code>is_holiday</code>	Binary indicator of a Czech public holiday.

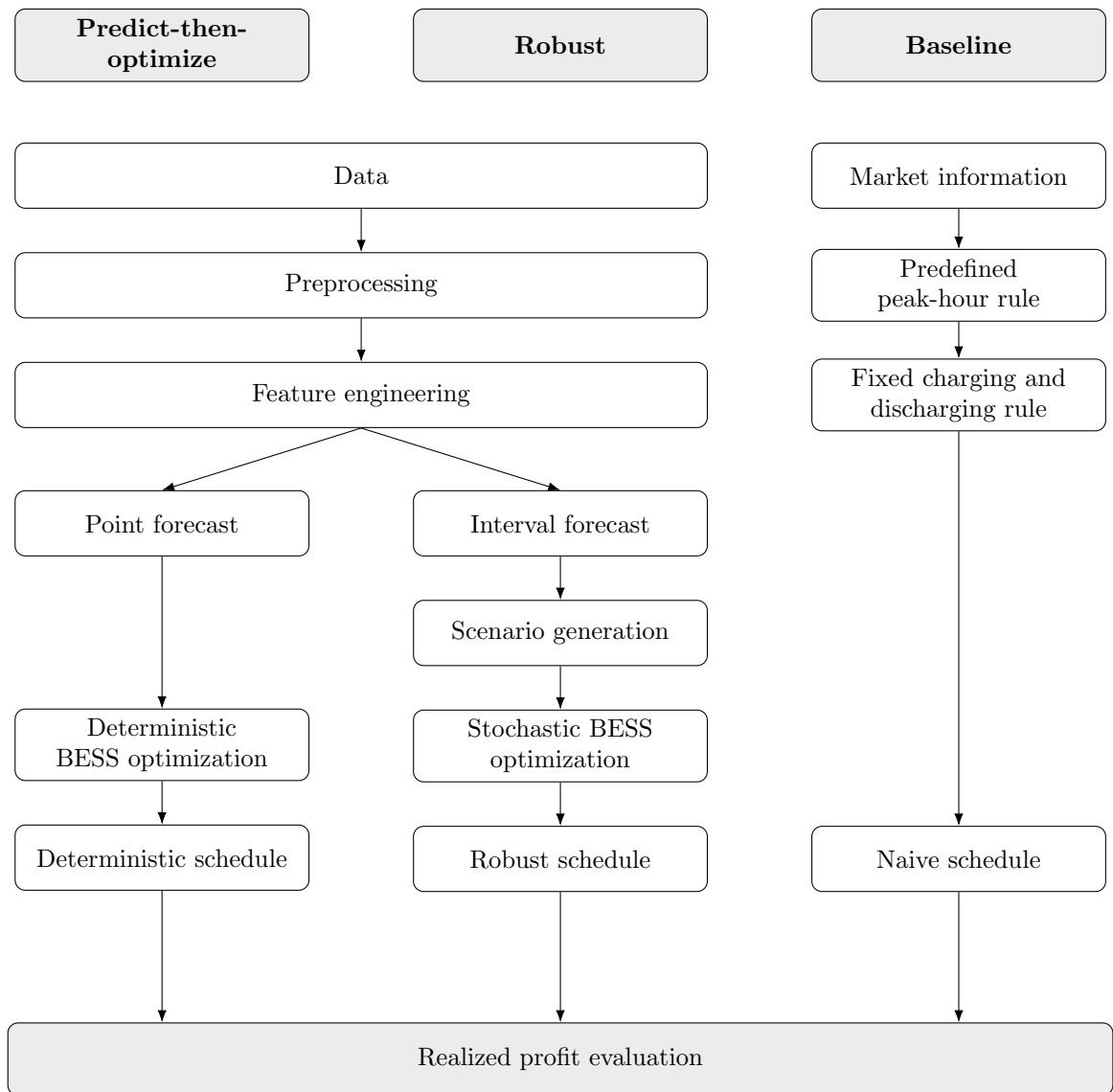
The variables `quarter_of_day_sin` and `quarter_of_day_cos` are cyclic transformations of the quarter-of-day index. They provide additional continuous information about the position within the daily cycle and help recognize that the end of one day is close to the beginning of the next day.

Forecasting and BESS optimization methodology

This chapter describes the complete decision pipeline used to transform market information and price forecasts into an operational Battery Energy Storage System schedule. The objective of this chapter is to describe the complete methodological framework used in this thesis. It explains all necessary steps, including day-ahead price forecasting, prediction interval construction, scenario generation, deterministic and stochastic BESS scheduling, and the ex-post evaluation of realized arbitrage profit.

Three scheduling strategies are considered. The first is a naive baseline based on predefined charging and discharging periods motivated by the typical structure of electricity prices or basic market information. The schedule is created using peak hour. The second is a deterministic predict-then-optimize approach, in which a point forecast is treated as the future price trajectory and a single optimal schedule is computed from it. The third is a stochastic approach that uses prediction intervals to generate price scenarios and then determines a schedule that explicitly accounts for uncertainty.

Figure 4.1 summarizes the three pipelines analyzed in this thesis. The first part of the chapter explains the forecasting stage, including both point and interval prediction. The second part presents the optimization layer, namely the deterministic model, scenario generation, the stochastic model, and the ex-post calculation of realized profit that is later used for the empirical comparison of all methods.



■ **Figure 4.1** Overview of the three scheduling pipelines.

4.1 Dataset specification

The experiments in this thesis are based on the forecasting dataset described in Chapter 3. The dataset contains both market variables obtained from external data sources and artificial features derived during preprocessing. To simulate the real use of the schedule creation pipeline, the data are split chronologically into training, calibration, and testing periods. Table 4.1 summarizes the time split used in the experiments in result chapter 5.

■ **Table 4.1** Overview of the dataset splits used in the experiments

Subset	Start	End (excluded)	Days covered
All data	2024-01-01	2025-09-01	609
Training data	2024-01-01	2025-04-01	517
Calibration data	2025-04-01	2025-07-01	91
Testing data	2025-07-01	2025-09-01	62

Note: The calibration dataset is included in the training period when calibration is not required.

The implementation is designed for the new 15-minute market time unit because the Czech day-ahead market switched from hourly to 15-minute trading intervals for delivery day 1 October 2025. For this reason, the scheduling framework is formulated at a 15-minute resolution.

The main experiments are evaluated on the summer 2025 testing period, because this period is expected to contain higher price variability and therefore greater arbitrage potential for BESS operation. Since the evaluated historical prices were still hourly, they were converted to 15-minute resolution by assigning the same hourly price to each of the four corresponding 15-minute intervals.

4.2 Model settings for forecasting

The forecasting models were trained with fixed hyperparameters in order to make the comparison between methods consistent. The selected parameter values were also chosen to keep the experiments computationally possible on a personal laptop. In the first approach, separate models were trained for the point forecast with the parameters shown in Table 4.2.

■ **Table 4.2** Main settings of the direct forecasting models

Model	Parameter	Meaning
Random Forest	<code>n_estimators = 100</code>	Number of decision trees used in the ensemble.
	<code>min_samples_leaf = 5</code>	Minimum number of observations required in each leaf; helps smooth predictions.
HGBR	<code>max_iter = 500</code>	Number of boosting iterations.
	<code>learning_rate = 0.05</code>	Controls the contribution of each new tree to the ensemble.
	<code>max_depth = 6</code>	Maximum depth of individual trees.
	<code>min_samples_leaf = 20</code>	Minimum number of observations in a leaf; reduces overfitting.
	<code>loss = squared_error</code>	Squared error is used for point prediction.
XGBoost	<code>n_estimators = 500</code>	Number of boosting trees.
	<code>learning_rate = 0.05</code>	Step size used when adding new trees.
	<code>max_depth = 6</code>	Maximum tree depth.
	<code>subsample = 0.9</code>	Fraction of training observations used for each tree.
	<code>colsample_bytree = 0.9</code>	Fraction of input features used for each tree.
	<code>loss = reg:squarederror</code>	Squared-error objective is used for forecasts
CatBoost	<code>iterations = 500</code>	Number of boosting iterations.
	<code>learning_rate = 0.05</code>	Controls how strongly each tree updates the model.
	<code>depth = 6</code>	Depth of the decision trees.
	<code>loss_function = RMSE</code>	RMSE is used for point prediction, while quantile loss is used for interval bounds.
	<code>cat_features</code>	Names of categorical or discrete variables

For interval price forecasting, the MAPIE framework was used to implement the EnbPI algorithm described in Section 2.4.2. In MAPIE, EnbPI is available through the `TimeSeriesRegressor` class [29]. This model requires a base estimator.

■ **Table 4.3** Main settings of the MAPIE EnbPI framework

Component	Parameter	Meaning
TimeSeriesRegressor	<code>method = enbpi</code>	Uses the Ensemble Batch Prediction Intervals method.
	<code>estimator = base_model</code>	Several models were tested, including Linear Regression, Random Forest, HGBR, XGBoost, and CatBoost.
	<code>n_resamplings = 35</code>	Number of bootstrap resamples used to construct the ensemble.
	<code>block_length = 96</code>	Length of each bootstrap block, corresponding to one complete day.
	<code>agg_function = mean</code>	Aggregates ensemble predictions by their mean.
	<code>confidence_level = 1-α</code>	Defines the nominal coverage level of the prediction interval.

Segmented Catboost

The `FourCatBoost`¹ is a segmented forecasting model [30] in which the training data are divided into several time-of-day blocks and a separate CatBoost model is trained for each block. Let q_t denote the quarter-of-day index of observation t , with $q_t \in \{0, \dots, 95\}$. The day is partitioned into four disjoint intervals

$$B_1 = [0, 32), \quad B_2 = [32, 52), \quad B_3 = [52, 76), \quad B_4 = [76, 96),$$

which approximately correspond to the periods 00:00–08:00, 08:00–13:00, 13:00–19:00, and 19:00–24:00. For each block B_k , a separate regression model $f_k(x)$ is trained only on observations satisfying $q_t \in B_k$. The final predictor can therefore be written as

$$\hat{y}_t = f_k(x_t), \quad \text{if } q_t \in B_k.$$

¹This is the implementation name used in the source code.

This means that the relationship between the input variables and the price is allowed to differ across different parts of the day. Such a decomposition is motivated by the electricity price forecasting literature, where hour-specific or interval-specific models are often used instead of a single global model for the whole day.

4.3 BESS scheduling strategies

Before applying optimization-based scheduling methods, two simple baseline strategies are defined. These baselines are used as reference cases for evaluating whether the forecasting. The first baseline is based on fixed peak hours, while the second baseline assumes that the optimal schedule from the previous day can be reused for the following day.

4.3.1 Peak-hour schedule

The peak-hour schedule is a simple rule-based strategy based on the typical daily structure of electricity prices. In this approach, the discharge period is fixed in advance and corresponds to the hours in which electricity prices are usually expected to be high. The battery is therefore charged before the selected peak-price period and discharged during the predefined peak hours.

In the implementation, the battery is charged during two fixed periods: 04:00–05:45 and 15:00–16:45. It is then discharged during the expected peak-price periods 07:00–08:45 and 18:00–19:45. This schedule was already shown as an illustrative example in Fig. 2.5 in Section 2.6.

This method does not use a price forecast or an optimization model. Instead, it relies only on the general assumption that electricity prices tend to follow a repeating intraday pattern. The peak-hour schedule is therefore useful as a naive benchmark, because it represents a simple operating rule that could be applied without complex forecasting or optimization.

4.3.2 Yesterday's optimal schedule

The second baseline is based on the assumption that the price pattern on day D is similar to the price pattern observed on day $D - 1$. Since the true prices for day $D - 1$ are already known when the schedule for day D is created, an optimal schedule can first be computed using the realized prices from the previous day. This schedule is then reused as the operating schedule for day D .

Formally, let $a_{D-1,t}^*$ denote the optimal action in time interval t obtained from the realized prices of day $D - 1$. The yesterday-based schedule assumes that the action for day D is given by

$$a_{D,t} = a_{D-1,t}^*.$$

This strategy is still simple, but requires deterministic schedule optimization 4.3.3. but it is more adaptive than the fixed peak-hour schedule because it uses the most recent observed market behavior.

4.3.3 Deterministic battery scheduling model

To compute the day-ahead charging and discharging schedule of the battery, a deterministic mixed-integer optimization model is used. The model assumes that **one price trajectory** is known in advance and finds the schedule with the highest possible arbitrage profit over one day.

Let $t = 0, \dots, T - 1$ denote the 15-minute intervals of the optimization horizon. For one day, $T = 96$. The decision variables are

$$c_t \geq 0, \quad d_t \geq 0, \quad SOC_t \geq 0, \quad z_t \in \{0, 1\},$$

where c_t is the charged energy in interval t , d_t is the discharged energy in interval t , SOC_t is the state of charge, and z_t is a binary variable that determines whether the battery is in charging or discharging mode.

The state of charge evolves according to

$$SOC_{t+1} = SOC_t + \eta^{\text{ch}} c_t - \frac{d_t}{\eta^{\text{dis}}},$$

where η^{ch} and η^{dis} are the charging and discharging efficiencies. The state of charge is limited by the battery capacity,

$$0 \leq SOC_t \leq \bar{E},$$

and the initial and final state of charge are fixed:

$$SOC_0 = E^{\text{init}}, \quad SOC_T = E^{\text{final}}.$$

The charging and discharging limits per 15-minute interval are

$$0 \leq c_t \leq \bar{C} z_t, \quad 0 \leq d_t \leq \bar{D} (1 - z_t),$$

where $\bar{C}$ and $\bar{D}$ are the maximum charge and discharge amounts per interval. The binary variable z_t prevents simultaneous charging and discharging.

To restrict excessive battery usage, the model also includes a daily cycle limit:

$$\sum_{t=0}^{T-1} c_t \leq N_{\text{cyc}} \bar{E}, \quad \sum_{t=0}^{T-1} d_t \leq N_{\text{cyc}} \bar{E},$$

where N_{cyc} is the maximum allowed number of equivalent full cycles per day.

The objective is to maximize the total profit:

$$\max \sum_{t=0}^{T-1} \left(\hat{y}_t d_t - \hat{y}_t c_t - \text{cost}^{\text{deg}}(c_t + d_t) \right),$$

where $cost^{\text{deg}}$ is the degradation or throughput cost per MWh. The first term represents revenue from discharging, the second term is the cost of charging, and the third term penalizes battery usage.

The complete optimization problem is therefore

$$\max \sum_{t=0}^{T-1} \left(\hat{y}_t d_t - \hat{y}_t c_t - cost^{\text{deg}}(c_t + d_t) \right)$$

subject to

$$\begin{aligned} SOC_{t+1} &= SOC_t + \eta^{\text{ch}} c_t - \frac{d_t}{\eta^{\text{dis}}}, \\ 0 &\leq e_t \leq \bar{E}, \\ SOC_0 &= E^{\text{init}}, \quad SOC_T = E^{\text{final}}, \\ 0 &\leq c_t \leq \bar{C} z_t, \quad 0 \leq d_t \leq \bar{D}(1 - z_t), \\ \sum_{t=0}^{T-1} c_t &\leq N_{\text{cyc}} \bar{E}, \quad \sum_{t=0}^{T-1} d_t \leq N_{\text{cyc}} \bar{E}, \\ z_t &\in \{0, 1\}. \end{aligned}$$

In the implementation, the model is formulated in Pyomo and solved using the HiGHS solver.

4.3.4 Stochastic battery scheduling model

The stochastic scheduling model extends the deterministic formulation by considering **multiple possible future price scenarios**. In contrast to the deterministic model, which optimizes the battery schedule for a single point forecast, the stochastic model evaluates the same day-ahead schedule under a set of price scenarios. For each scenario $s = 1, \dots, S$, the realized profit of the schedule is defined as

$$\Pi_s = \sum_{t=0}^{T-1} \left(\hat{y}_{s,t} d_t - \hat{y}_{s,t} c_t - cost^{\text{deg}}(c_t + d_t) \right),$$

where $\hat{y}_{s,t}$ is the electricity price in scenario s and time interval t . The objective is based on Conditional Value-at-Risk (CVaR), which penalizes poor performance in adverse scenarios.

The scenario loss is defined as the negative profit,

$$L_s = -\Pi_s.$$

For a confidence level $\beta \in (0, 1)$, the CVaR of the loss distribution is introduced using the auxiliary variables ξ and u_s [31]:

$$\text{CVaR}_\beta(L) = \xi + \frac{1}{(1 - \beta)S} \sum_{s=1}^S u_s.$$

The variable ξ represents the loss threshold from which the worst scenarios start to be penalized. The auxiliary variable u_s measures how much the loss in scenario s exceeds this threshold. Therefore, it is defined through the relation

$$u_s = \max(L_s - \xi, 0).$$

If the loss L_s is below the threshold ξ , then $u_s = 0$ and the scenario does not contribute to the risk penalty. If the loss is larger than ξ , then u_s becomes positive and increases the CVaR value. In this way, the CVaR term penalizes schedules that perform poorly in adverse scenarios, while schedules with lower losses in the tail of the distribution receive a smaller penalty.

The risk-averse stochastic scheduling model maximizes the expected profit while penalizing downside risk:

$$\max \left[\frac{1}{S} \sum_{s=1}^S \Pi_s - \lambda \left(\xi + \frac{1}{(1-\beta)S} \sum_{s=1}^S u_s \right) \right],$$

where $\lambda \geq 0$ is a risk-aversion parameter. When $\lambda = 0$, the model reduces to a risk-neutral stochastic model that maximizes expected profit. For larger values of λ , the optimizer gives more importance to avoiding low-profit outcomes in adverse price scenarios.

The model is subject to the same battery constraints as the deterministic model:

$$\begin{aligned} SOC_{t+1} &= SOC_t + \eta^{\text{ch}} c_t - \frac{d_t}{\eta^{\text{dis}}}, \\ 0 &\leq SOC_t \leq \bar{E}, \\ SOC_0 &= E^{\text{init}}, \quad SOC_T = E^{\text{final}}, \\ 0 &\leq c_t \leq \bar{C}z_t, \quad 0 \leq d_t \leq \bar{D}(1 - z_t), \\ \sum_{t=0}^{T-1} c_t &\leq N_{\text{cyc}} \bar{E}, \quad \sum_{t=0}^{T-1} d_t \leq N_{\text{cyc}} \bar{E}, \\ z_t &\in \{0, 1\}, \end{aligned}$$

$$u_s \geq L_s - \xi, \quad u_s \geq 0.$$

This formulation remains a mixed-integer linear programming problem, because the profit expressions and the CVaR constraints are linear in the decision variables. The CVaR confidence level and the risk-aversion parameter were fixed to $\beta = 0.95$ and $\lambda = 0.1$, in all stochastic scheduling experiments.

Interval-based Monte Carlo scenario generation

To convert interval forecasts into full-day price scenarios, a Monte Carlo sampling procedure is used. For each time interval t , the lower and upper prediction bounds are denoted by y_t^L and y_t^U . These bounds define the range of possible prices for that interval.

For each scenario $s = 1, \dots, S$ and each time interval t , a random price is sampled from a uniform distribution between the lower and upper prediction bounds:

$$y_{t,s} \sim \mathcal{U}(y_t^L, y_t^U).$$

In this way, each scenario represents one possible full-day price trajectory consistent with the predicted interval bounds. Repeating this sampling procedure for all scenarios produces a set of price paths that can be used as input to the stochastic BESS scheduling model.

Residual bootstrap scenario generation

The residual bootstrap method generates stochastic price scenarios by reusing historical forecast errors. Let y_t denote the observed electricity price and $\hat{y}_t$ the corresponding point forecast. The forecast residual at time t is defined as

$$\varepsilon_t = y_t - \hat{y}_t.$$

Historical residuals are therefore grouped into daily residual vectors,

$$\boldsymbol{\varepsilon}_d = (\varepsilon_{d,1}, \varepsilon_{d,2}, \dots, \varepsilon_{d,T}),$$

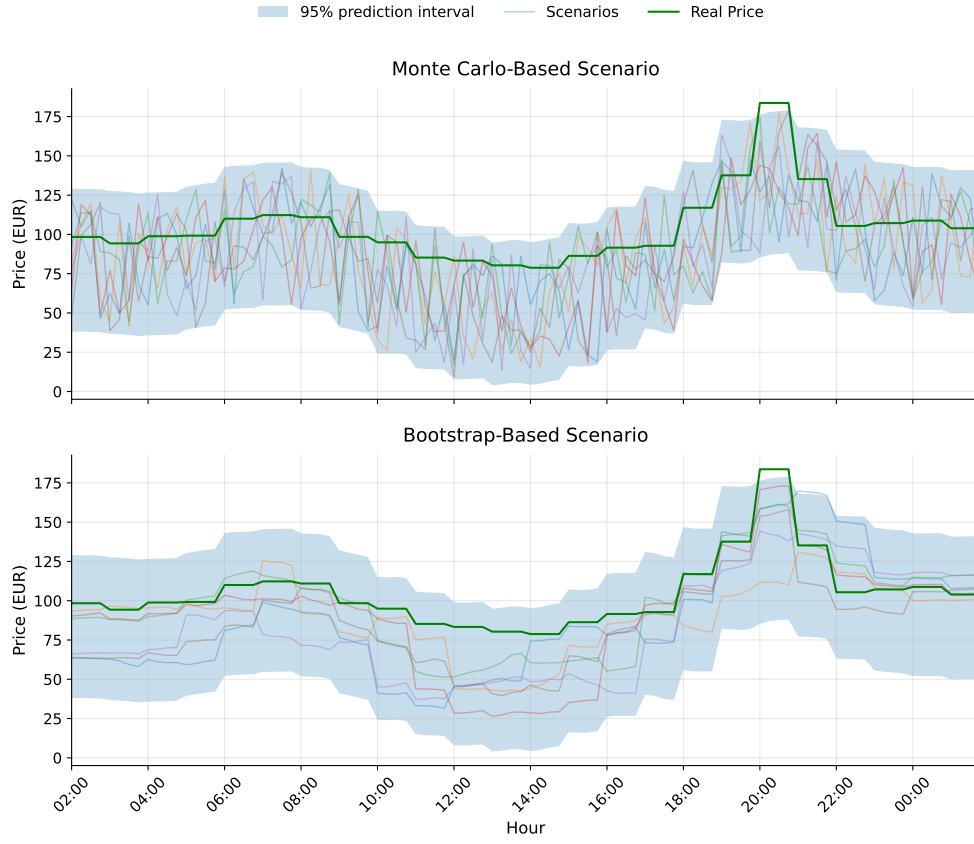
where $d = 1, \dots, N$ indexes complete historical days. In this way, a residual library is obtained:

$$R = [\boldsymbol{\varepsilon}_1, \boldsymbol{\varepsilon}_2, \dots, \boldsymbol{\varepsilon}_N]^\top \in \mathbb{R}^{N \times T}.$$

To generate scenarios for a new target day $\hat{\mathbf{y}} \in \mathbb{R}^T$ denote the point forecast for that day. A bootstrap scenario is obtained by sampling one complete historical residual vector from the residual library and adding it to the point forecast. Scenario s is constructed as

$$\mathbf{y}^{(s)} = \hat{\mathbf{y}} + \boldsymbol{\varepsilon}_{i,:}$$

where $i, :$ is the index of a randomly selected historical day from library. Figure 4.2 illustrates several randomly selected scenarios produced by this method and compares them with the Monte Carlo approach.



■ **Figure 4.2** Comparison of the two scenario generation methods for the day 2025-07-25.

4.3.5 Know prices schedule optimization

The last schedule used in this thesis as benchmark is the **perfect-foresight schedule**. It serves as a theoretical benchmark for the best possible performance of the considered BESS model. In this case, the battery schedule is obtained using the deterministic scheduling model described in Section 4.3.3, but the input prices are the realized electricity prices y of the same delivery day.

This schedule is not realistic in practical operation, because it assumes that the electricity prices of the whole delivery day are known before the schedule is created. However, the perfect-foresight schedule is useful as a theoretical benchmark. It shows the maximum profit that could be achieved by the considered BESS if the future prices were known exactly.

The profit obtained from this schedule is referred to as the **perfect-foresight profit**. It represents the best possible result under the selected BESS parameters and is used as an upper bound for evaluating all forecast-based scheduling strategies.

4.4 Evaluation metrics

The proposed scheduling pipeline is evaluated at several stages. First, the quality of point price forecasts is assessed using standard forecasting error metrics. Second, the quality of prediction intervals is evaluated in terms of their reliability and sharpness. Finally, the resulting BESS schedules are assessed using economic and operational metrics. This separation is important because a forecasting model with good statistical accuracy does not necessarily lead to the most profitable battery schedule. Therefore, both forecasting performance and realized scheduling performance are evaluated.

4.4.1 Point prediction metrics

Point forecasts are evaluated using RMSE, MAE, wMAPE, and R^2 . The first two metrics are scale-dependent and directly interpretable in the original price unit, EUR/MWh. MAE is defined as

$$\text{MAE} = \frac{1}{N} \sum_{t=1}^N |y_t - \hat{y}_t|,$$

where y_t is the true electricity price and $\hat{y}_t$ is its prediction. MAE is included because it measures the average absolute forecasting error and is less sensitive to extreme values than squared-error metrics.

RMSE is defined as

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{t=1}^N (y_t - \hat{y}_t)^2}.$$

In contrast to MAE, RMSE penalizes large errors more strongly. This is relevant for electricity prices, where occasional spikes can have a substantial impact on the quality of a forecast and on downstream trading decisions.

As a relative error measure, the weighted mean absolute percentage error (wMAPE) is also reported:

$$\text{wMAPE} = 100 \cdot \frac{\sum_{t=1}^N |y_t - \hat{y}_t|}{\sum_{t=1}^N |y_t|}.$$

This metric is preferred over the standard MAPE because the latter becomes unstable when the true values are very small. In electricity markets, this is important because prices may approach zero and standard percentage errors can then become misleading.

Finally, the coefficient of determination is used as a complementary goodness-of-fit statistic:

$$R^2 = 1 - \frac{\sum_{t=1}^N (y_t - \hat{y}_t)^2}{\sum_{t=1}^N (y_t - \bar{y})^2},$$

where $\bar{y}$ is the mean of the true prices. While R^2 is not the primary metric for operational decision-making, it is useful as a summary of how much variability in the observed prices is explained by the model.

4.4.2 Prediction interval metrics

Prediction intervals are evaluated using coverage, average interval width, and the Winkler score [32]. $[y_t^L, y_t^U]$ denote the lower and upper bound of a $(1 - \alpha)$ prediction interval for time step t . These metrics are also used in the MAPIE framework for assessing conformal prediction performance.

Coverage measures how often the true price falls inside the predicted interval:

$$\text{Coverage} = \frac{1}{N} \sum_{t=1}^N \mathbf{1}(y_t^L \leq y_t \leq y_t^U)$$

A well-calibrated method should achieve empirical coverage close to the nominal level $1 - \alpha$.

The average interval width is defined as

$$\text{Width} = \frac{1}{N} \sum_{t=1}^N (y_t^U - y_t^L).$$

Narrower intervals are preferred, provided that the required coverage is maintained. To jointly assess interval sharpness and validity, the Winkler score is used:

$$W_t = \begin{cases} (y_t^U - y_t^L), & \text{if } (y_t^L \leq y_t \leq y_t^U), \\ (y_t^U - y_t^L) + \frac{2}{\alpha}(y_t^L - y_t), & \text{if } y_t < y_t^L, \\ (y_t^U - y_t^L) + \frac{2}{\alpha}(y_t - y_t^U), & \text{if } y_t > y_t^U. \end{cases}$$

The final score is obtained as the average over all time steps,

$$\text{Winkler} = \frac{1}{N} \sum_{t=1}^N W_t.$$

Lower values are better. The Winkler score rewards narrow intervals when the observation is covered, but penalizes intervals that miss the true value.

4.4.3 Schedule evaluation metrics

Since the ultimate goal of the thesis is battery trading rather than forecasting alone, the schedules are also evaluated using economic metrics. The main metric is the total cumulative realized profit, computed by applying the schedule to the true market prices:

$$\Pi = \sum_{t \in T} (y_t d_t - y_t c_t - \text{cost}^{\text{thr}}(c_t + d_t)),$$

where y_t is the realized electricity price, d_t is discharged energy, c_t is charged energy, and $cost^{\text{thr}}$ is the throughput cost coefficient.

To compare the operational behavior of a forecast-driven schedule with the perfect-foresight benchmark, a true schedule match metric is also used. Let a_t denote the action implied by the evaluated schedule and a_t^* the action implied by the true-price optimal schedule. Then

$$\text{Match} = \frac{1}{|T|} \sum_{t \in T} \mathbf{1}\{a_t = a_t^*\}.$$

This metric expresses how often the forecast-based schedule chooses the same action as the optimal schedule under true prices.

Finally, the economic loss relative to perfect foresight is defined as

$$\text{Loss} = \Pi^* - \Pi,$$

where Π^* is the cumulative profit obtained by optimizing directly on the true prices. This metric quantifies the value lost due to forecast uncertainty and imperfect scheduling decisions.

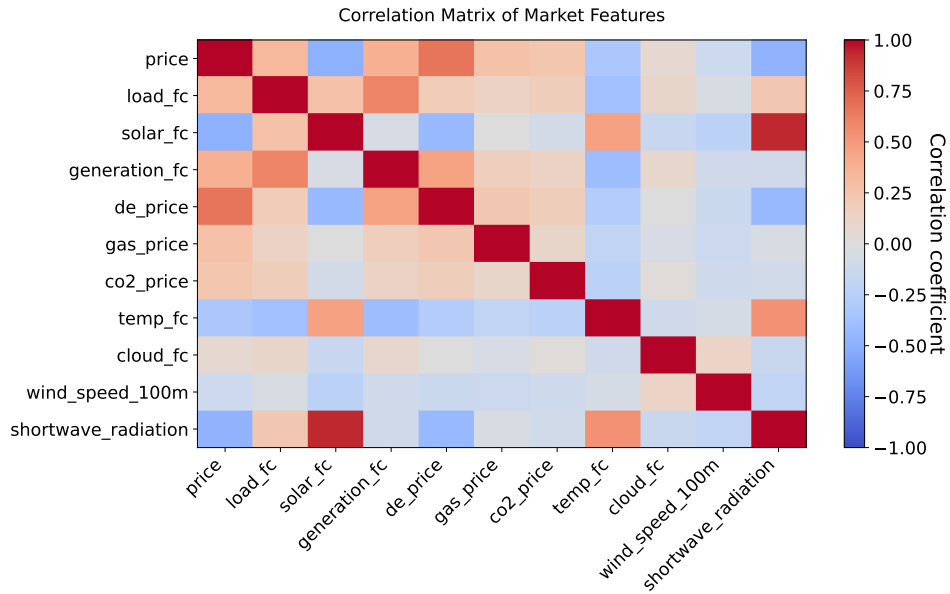
Results and discussion

This chapter presents and discusses the experimental results of the proposed BESS scheduling pipeline. First, the performance of point and interval forecasting models is evaluated, including the effect of daily model updating. Finally, the resulting BESS schedules are compared in terms of realized profit and similarity to the perfect-foresight schedule. The chapter also includes an additional winter-period experiment to show how the available arbitrage opportunity changes under different seasonal market conditions.

5.1 Characteristics of used features

Before presenting the forecasting results, the relationships between the selected input variables are examined using the Pearson correlation coefficient. The correlation matrix is used as an exploratory tool to identify basic linear dependencies between the variables. This is useful before model evaluation, because it provides an initial overview of how the input features are related to the target variable and whether some variables contain similar information.

Fig. 5.1 shows the correlation matrix for the market, system, and weather variables. The observed relationships are consistent with the expected behaviour of electricity prices. Renewable-related variables, such as `solar_fc`, `shortwave_radiation`, and partly also `temp_fc`, show a negative relationship with the electricity price. In contrast, several cost related variables show a positive relationship with prices, which is also expected, since higher load or higher fuel and emission costs tend to increase the market-clearing price.



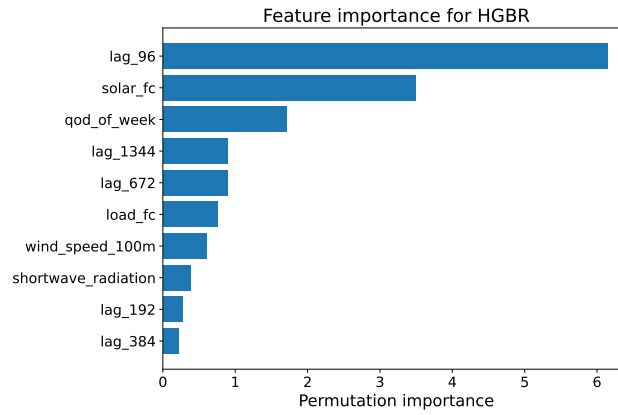
■ **Figure 5.1** Pearson correlation matrix of market features.

It should be emphasized that the correlation matrix is only an exploratory diagnostic. It describes linear associations in the data, but it does not prove causal relationships between the features and the electricity price.

5.1.1 Feature importance comparison

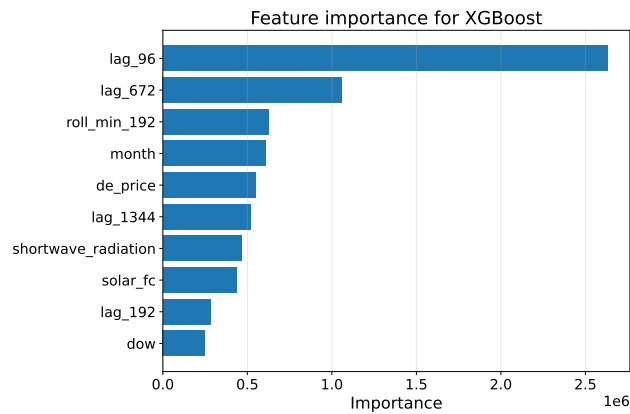
This analysis indicates which variables had the strongest influence on the fitted forecasting models. The purpose of the comparison is not to directly compare the absolute importance values across models, because each method computes feature importance differently. Instead, the aim is to identify which features are consistently preferred by the individual models.

For HGBR, **Permutation Feature Importance** was used [33]. This method evaluates how important a feature is by first measuring the model performance on the original data and then randomly shuffling the values of one feature at a time. If the model performance decreases strongly after shuffling a feature, it means that the model relied on this feature for its predictions. The results are shown in Fig. 5.2.



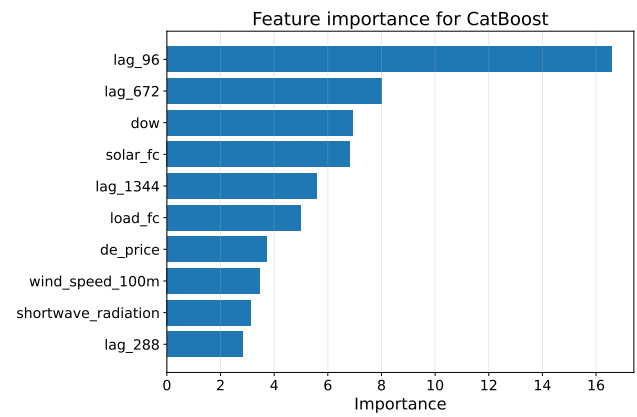
■ **Figure 5.2** Permutation feature importance of the HGBR model.

For the XGBoost model, feature importance was evaluated using the build-in `get_score` method with the gain criterion [34]. In this setting, gain describes how much a feature improves the model objective when it is used for a split in a decision tree. A higher gain value means that splits based on that feature contributed more strongly to reducing the prediction error. Since the reported value is averaged over all splits where the feature appears, the gain criterion reflects the average usefulness of a feature when it is selected by the model. Shown in Fig. 5.3.

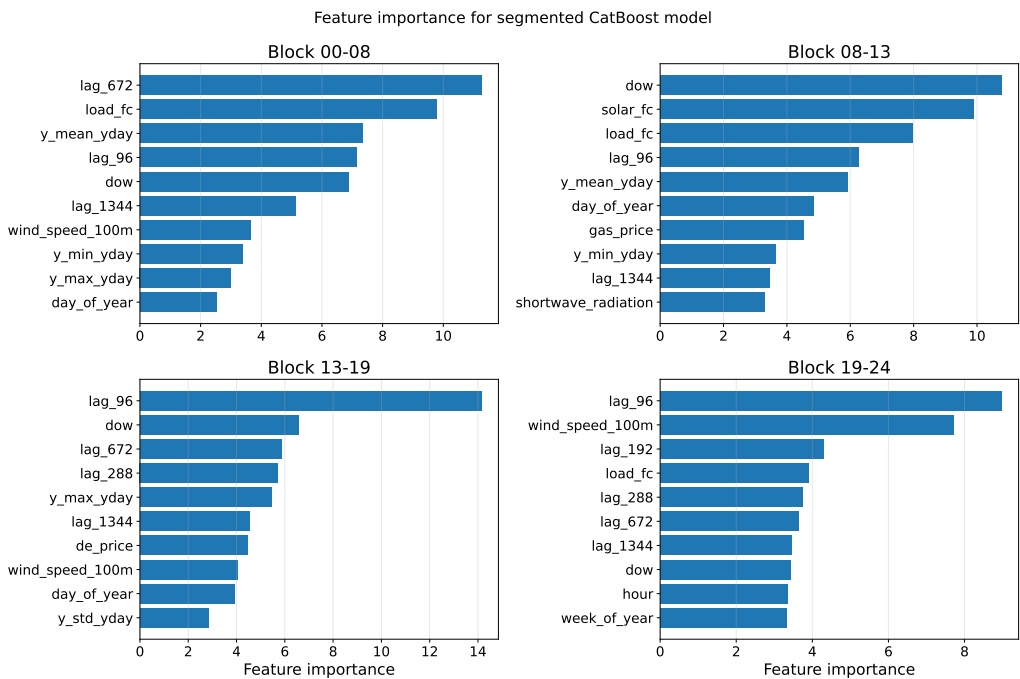


■ **Figure 5.3** Feature importance of the XGBoost model based on gain.

For CatBoost, the built-in `get_feature_importance` method was used [35]. This method calculates how strongly individual input features contribute to the predictions of the trained model. In tree-based models, a feature is considered important when it is frequently used in splits that meaningfully affect the model output. The resulting importance values are shown in Fig. 5.4.



■ **Figure 5.4** Feature importance of the CatBoost model.



■ **Figure 5.5** Feature importance of the segmented CatBoost model for individual blocks.

Finally, Fig. 5.5 shows feature importance for the segmented CatBoost model for each time-of-day block.

This analysis should not be interpreted as evidence of causality. It provides only a model-based overview of which features contributed most strongly to the predictions of a given model. A useful example is the segmented CatBoost model, where the block trained from 08:00 to 13:00 assigns relatively high importance to `solar_fc`. Which is reasonable, since solar generation is most relevant during daylight hours. A similar pattern can also be seen for `wind_speed_100m`.

5.2 Point forecasting results

To examine which model provides best point forecast, models were evaluated in two settings: without model updating and with daily model updating. In the setting without updating, each model is trained once on the initial training dataset and then used to predict the whole testing period, Table 5.1.

In the daily updating setting, the model first predicts prices for one complete delivery day. After the true prices for that day become available, this day is added to the training dataset and the model is retrained before predicting the next day. In this way, the model can gradually incorporate the most recent market information, Table and 5.2.

■ **Table 5.1** Point forecasting results without model updating

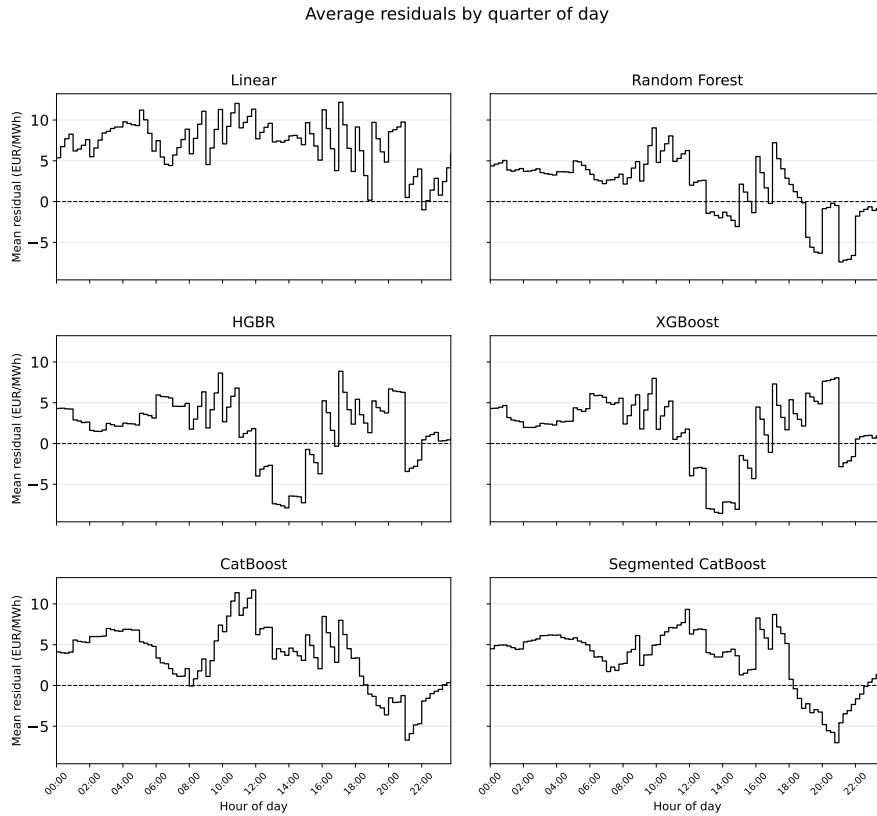
Model	wMAPE (%)	MAE	RMSE	R ²
Linear	20.36	17.13	24.57	0.68
Random Forest	17.94	15.09	25.76	0.65
HGBR	17.53	14.75	24.23	0.69
XGBoost	17.34	14.59	23.76	0.70
CatBoost	20.33	17.10	25.13	0.67
Segmented CatBoost	18.98	15.97	23.99	0.70

■ **Table 5.2** Point forecasting results with daily model updating

Model	wMAPE (%)	MAE	RMSE	R ²
Linear	19.62	16.50	24.13	0.69
Random Forest	17.73	14.91	25.70	0.65
HGBR	17.26	14.52	25.20	0.67
XGBoost	17.04	14.34	23.98	0.70
CatBoost	17.01	14.31	24.05	0.70
Segmented CatBoost	17.90	15.06	23.78	0.70

Fig. 5.6 provides a more detailed view of how the forecast errors change during the day for models with daily updating. The residuals are computed as $r_t = y_t - \hat{y}_t$, and then averaged separately for each 15-minute interval of the day. Positive values therefore indicate underprediction, while negative values indicate overprediction.

The figure 5.6 shows that the linear model tends to underpredict prices during most parts of the day. For most of the other models, the average residuals are relatively small and stable during the early morning hours. Larger deviations appear later in the day, especially around periods with stronger price dynamics, such as midday and evening peak hours. This suggests that these parts of the day are more difficult to forecast accurately.

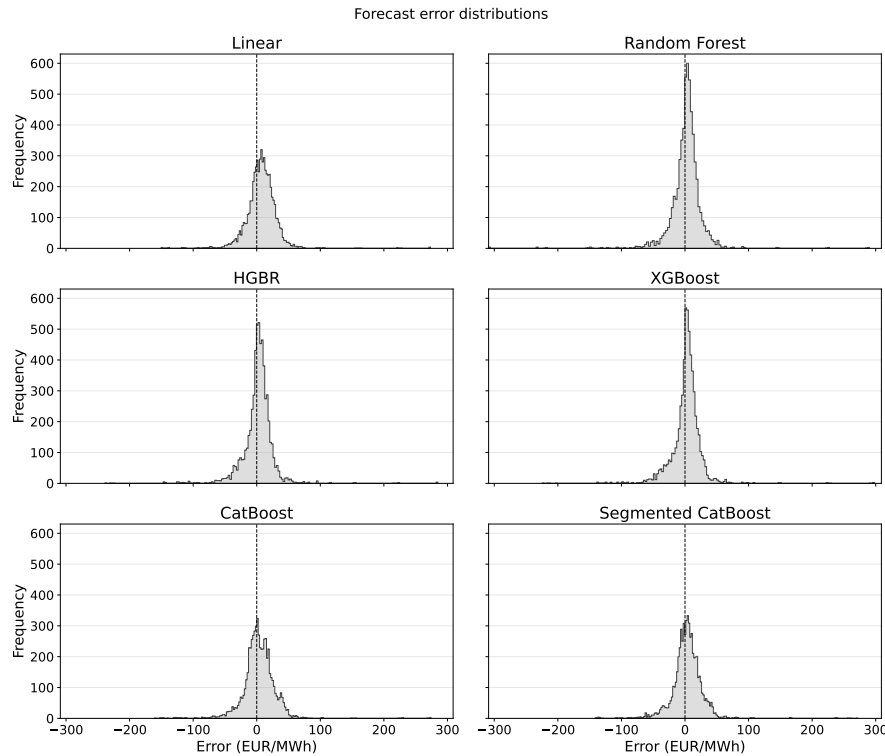


■ **Figure 5.6** Average forecast residuals by 15-minute interval of the day.

Fig. 5.7 shows the distribution of forecast errors for the models with daily updating. The error is computed as $y_t - \hat{y}_t$, so positive values indicate that the model underpredicted the realized price, while negative values indicate overprediction.

The histograms show that most errors are concentrated close to zero, which means that large forecast errors occur less frequently than small ones. At the same time, the error distributions are not perfectly symmetric. The Linear,

XGBoost, Random Forest and HGBR models show a slight shift toward positive errors, suggesting that they tend to underpredict prices in some periods. Random Forest and CatBoost appear more centered around zero, although they still contain occasional larger errors. Overall, the plots indicate that the models do not have a strong systematic bias, but some differences in the shape and spread of the error distributions remain visible.



■ **Figure 5.7** Distribution of forecast errors for models with daily updating. Positive errors indicate underprediction, while negative errors indicate overprediction.

5.3 Interval forecasting results

Since daily updating generally improved the point forecasting results, the interval forecasting results are reported only for the updated models. Table 5.3 presents the results for the quantile-based prediction intervals described in Section 2.4.1. The miscoverage level was set to $\alpha = 0.05$, which corresponds to a nominal coverage level of 95%.

■ **Table 5.3** Prediction interval results with daily model updating

Model	Coverage (%)	Width	Winkler score
Linear	97.80	110.27	142.24
Random Forest	83.35	54.43	154.98
HGBR	95.46	103.12	137.26
XGBoost	95.75	101.81	138.25
CatBoost	90.15	62.28	119.67
Segmented CatBoost	86.72	58.71	140.50

The result Table 5.3 show a clear trade-off between coverage and interval width. The Linear model produces the highest coverage, but also the widest intervals. HGBR and XGBoost achieve coverage close to the desired 95% level, although their intervals remain relatively wide. CatBoost obtains the lowest Winkler score, which indicates a good balance between interval width and missed observations. Random Forest produces the narrowest intervals, but its coverage is only 83.35%.

Table 5.4 reports the corresponding results obtained with the EnbPI method.

■ **Table 5.4** Prediction interval results for base models with the EnbPI method

Model	Coverage (%)	Width	Winkler score
Linear	97.01	98.16	136.12
Random Forest	96.24	97.25	147.75
HGBR	96.00	89.90	135.97
XGBoost	96.10	88.87	134.64
CatBoost	96.25	88.02	132.12
Segmented CatBoost	96.02	91.20	131.17

Compared with the direct quantile-based intervals, EnbPI provides much more stable empirical coverage across all tested base models. All models achieve coverage close to or above the nominal 95% level. CatBoost achieves the narrowest intervals, while the segmented CatBoost model obtains the lowest Winkler score.

5.4 Battery scheduling results

The most important results concern how the quality of the forecasts affects the resulting battery schedules and their financial performance. Once the price predictions were obtained, corresponding battery schedules were created and subsequently evaluated. The same BESS configuration was used in all scheduling experiments, and its main parameters are summarized in Table 5.5.

The cumulative profit was evaluated over the 62-day testing period in summer 2025, as specified in Table 4.1.

■ **Table 5.5** Selected BESS parameters used in the scheduling experiments

Parameter	Symbol	Value
Energy capacity	$\bar{E}$	100 MWh
Maximum charge per 15-minute interval	$\bar{C}$	12.5 MWh
Maximum discharge per 15-minute interval	$\bar{D}$	12.5 MWh
Initial state of charge	E^{init}	0 MWh
Final state of charge	E^{final}	0 MWh
Charging efficiency	η^{ch}	0.95
Discharging efficiency	η^{dis}	0.95
Degradation cost	cost^{deg}	5 EUR/MWh
Maximum cycles per day	N^{cyc}	2

5.4.1 Predict-then-optimize battery scheduling

Table 5.6 presents the results of the deterministic predict-then-optimize approach with daily model updating.

The column **Profit (%)** reports the realized profit as a percentage of the perfect-foresight benchmark, which equals 662,215.55 EUR for the evaluated period. The column **Match Rate (%)** measures how often the forecast-based schedule selected the same action as the schedule optimized using the realized prices.

■ **Table 5.6** Schedule evaluation results for the predict-then-optimize approach

Model	Profit (%)	Match rate (%)
Linear	91.39	90.07
Random Forest	86.45	88.37
HGBR	89.47	89.94
XGBoost	89.69	89.67
CatBoost	91.72	90.41
Segmented CatBoost	92.27	90.36

5.4.2 Robust scheduling results

The robust scheduling approach that uses EnbPI prediction intervals to generate multiple possible price scenarios. These scenarios are then used in the stochastic optimization model, which creates one battery schedule that performs well across the considered uncertainty set. Table 5.7 summarizes the

results obtained with Monte Carlo scenario generation. The table also includes the bootstrap-based result for the segmented CatBoost model.

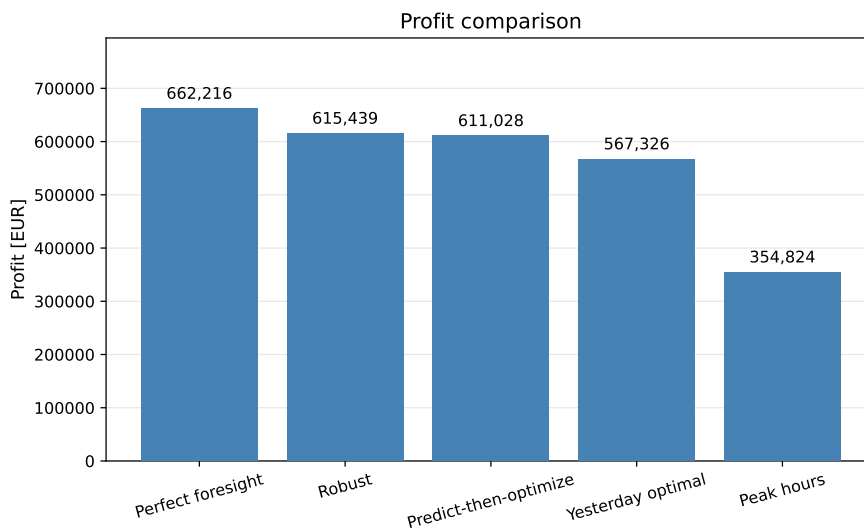
■ **Table 5.7** Robust schedule evaluation results with EnbPI-based scenario generation

Model	Profit (%)	Match Rate (%)
Linear	91.17	89.88
Random Forest	91.34	90.46
HGBR	92.26	90.76
XGBoost	92.69	90.98
CatBoost	92.56	90.86
Segmented CatBoost	92.94	90.89
Segmented CatBoost Bootstrap	92.87	91.00

5.5 Comparison of scheduling strategies

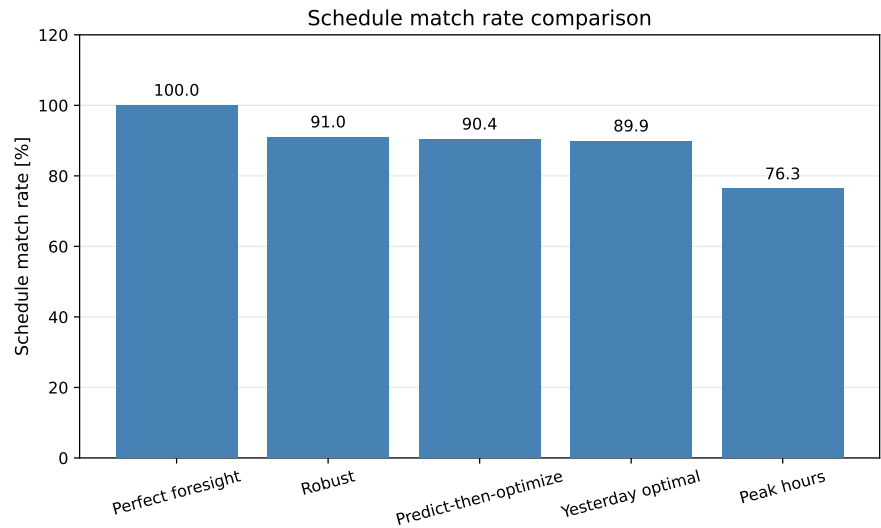
The previous sections evaluated the predict-then-optimize and robust scheduling approaches separately. This section compares all implemented scheduling strategies together, including the baseline schedules and the perfect-foresight benchmark.

Fig. 5.8 shows the realized profit in EUR for each implemented scheduling strategy. This comparison shows how much value is lost due to forecast uncertainty and imperfect scheduling decisions.



■ **Figure 5.8** Realized profit of the implemented BESS scheduling strategies.

Fig. 5.9 compares the true schedule match rate. Strategy may have a higher match rate but still achieve a lower profit if it makes incorrect decisions during the most valuable price periods.



■ **Figure 5.9** True schedule match rate of the implemented BESS scheduling strategies.

The robust and predict-then-optimize strategies achieve very similar results. In both cases, the best-performing schedules were obtained when the segmented CatBoost model was used as the forecasting model. The comparison also shows that optimization-based scheduling clearly improves over simple baseline strategies.

To illustrate the operational behavior of the best scheduling approaches, Fig. 5.10 shows the average charge and discharge schedule together with the corresponding average state of charge. The figure helps explain how the battery is typically used during the day and compare them with perfect-foresight schedule.

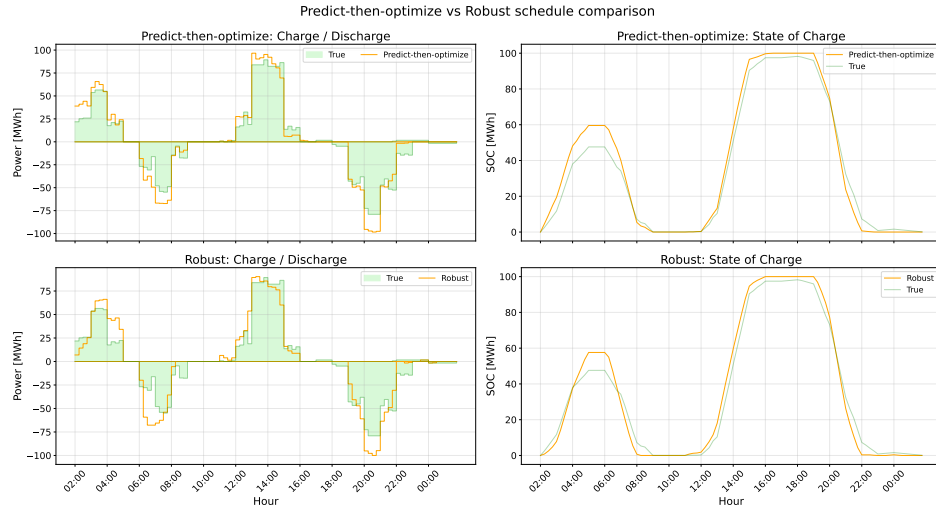


Figure 5.10 Average BESS schedule and corresponding state of charge for the selected scheduling strategy.

5.6 Winter-period scheduling experiment

To examine whether the profitability of BESS arbitrage depends also on the selected season, an additional experiment was performed on a winter testing period. The main experiment in this thesis uses summer data, where stronger intraday price variability and higher solar-related price fluctuations are expected. In contrast, the purpose of the winter experiment is to evaluate whether the arbitrage opportunity is lower when the daily price spread is less favorable for battery operation.

The same forecasting and scheduling pipeline was used as in the main experiment. Only the testing period was changed. The winter test period starts on 1 December 2025 and ends before 2 February 2026. The corresponding data split is shown in Table 5.8.

Table 5.8 Overview of the dataset split used for the winter experiment

Subset	Start	End (excluded)	Days covered
All data	2024-01-01	2026-02-02	763
Training data	2024-01-01	2025-09-01	609
Testing data	2025-12-01	2026-02-02	62

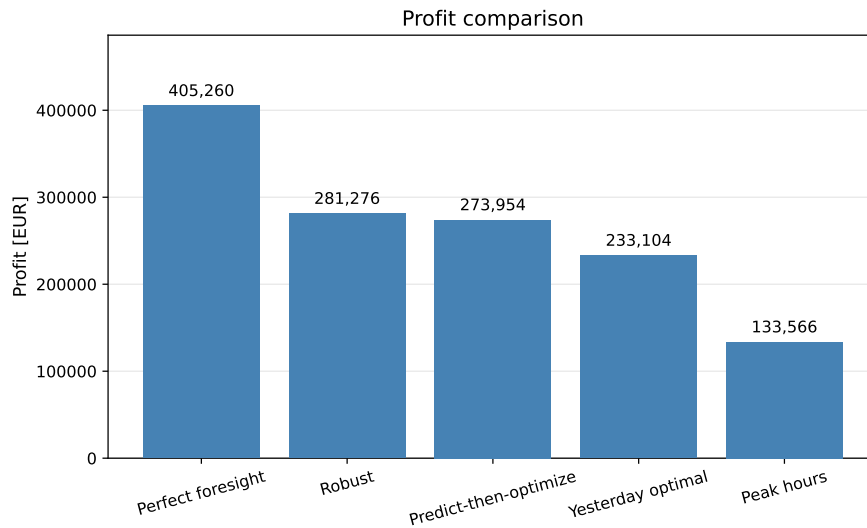
Note: The calibration dataset is included in the training period.

Table 5.9 summarizes the schedule evaluation results for the winter period. For both the predict-then-optimize and robust approaches, the segmented Cat-Boost model was used, as it achieved the best scheduling performance in the main summer experiment.

■ **Table 5.9** Schedule evaluation results for the winter testing period

Strategy	Profit (EUR)	Profit (%)	Match Rate(%)
Peak-hour	133,566	32.96	78.67
Yesterday's optimal	233,104	57.52	83.95
Predict-then-optimize	273,954	67.60	85.69
Robust	281,276	69.41	84.49
Perfect foresight	405,260	100.00	100.00

Fig. 5.11 shows the realized profit of the evaluated scheduling strategies during the winter testing period. This comparison is useful because the absolute level of achievable profit reflects the available arbitrage opportunity in the selected period. If the perfect-foresight profit and the forecast-based profits are lower than in the summer experiment, this indicates that the winter period provided less profitable price spreads for BESS operation.



■ **Figure 5.11** Realized profit of the implemented BESS scheduling strategies during the winter testing period.

The winter experiment provides an additional robustness check for the proposed pipeline. While the main summer experiment evaluates the framework under conditions with stronger expected price variability, the winter experiment shows how the same scheduling methods perform when the arbitrage opportunity is lower. The comparison therefore highlights that the profitability of BESS operation is not determined only by the forecasting and optimization method, but also by the underlying market conditions in the evaluated period.

Chapter 6

Conclusion

This chapter summarizes the main results of this thesis and evaluates the extent to which the stated objectives were achieved. It highlights the practical contribution, discusses its main limitations, and outlines possible directions for future research.

The first objective of the thesis was to acquire and preprocess historical market data relevant for the Czech day-ahead market. This goal was fulfilled by constructing a structured dataset containing electricity prices, market variables, weather forecasts, and additional engineered features.

The second objective was to develop forecasting models for Czech day-ahead electricity prices. Several models were implemented and compared. The results showed that daily model updating generally improved several forecasting metrics, especially wMAPE and MAE.

The next objective was to implement a baseline predict-then-optimize framework and a robust alternative that explicitly considers uncertainty. Both of these goals were achieved. The predict-then-optimize formulation used point forecasts as direct inputs for battery scheduling, while the robust formulation used prediction intervals to generate scenarios for stochastic optimization.

Both approaches were benchmarked against simple reference strategies, namely the fixed peak-hour schedule and the yesterday-optimal schedule, as well as against the perfect-foresight schedule, which represents the theoretical upper bound. An interesting result was the relatively strong performance of the yesterday-optimal schedule. This strategy does not require a price forecast, it still achieved a competitive profit compared with the forecast-based approaches.

The results from experiments on historical market data showed that all optimization-based approaches clearly outperformed the simple schedule. The deterministic predict-then-optimize method already provided strong performance. The robust approach achieved slightly higher profits than the deter-

ministic one, which means that incorporating uncertainty can improve battery scheduling decisions.

From a practical perspective, the thesis demonstrates a complete pipeline for BESS scheduling. An important outcome of the work is the full implementation, which is provided in `src/BESS_schedule.ipynb`. The framework includes data preprocessing, point and interval forecasting, deterministic and stochastic optimization, and simulation-based evaluation.

The thesis also has several limitations. The battery model remains simplified, and some of the considered models could not be fully used because of computational constraints.

Overall, the thesis showed that combining electricity price forecasting with optimization-based BESS scheduling is a feasible and practically relevant approach for the Czech electricity market. While the robust method achieved the best overall results, the deterministic predict-then-optimize framework also performed competitively, which indicates that even simpler scheduling strategies can provide strong results when supported by suitable forecasting models.

In addition, an interactive overview of the experimental results was created and is available at <https://bessschedule-production.up.railway.app/>. This overview allows the results from the experiments to be accessed and compared interactively.

6.1 Future work

Future work should mainly focus on addressing the identified limitations of the current framework. In particular, the battery model could be extended so that the simulations better reflect real operating conditions.

In addition to the day-ahead market, BESS can also be used in other electricity market. These include the intraday market, where storage can react to updated price signals and forecast errors closer to real time.

Appendix A

Interactive Result Overview

As a supplementary output of this thesis, an interactive web application was created to provide a quick overview of the experimental results. The application is not intended to explain or interpret the results. Its purpose is to make it possible to inspect different testing days, compare selected BESS schedules, and view their corresponding performance values in an interactive form.

The application is available at:

<https://bessschedule-production.up.railway.app/>

Fig. A.1 shows the page of the application.



Figure A.1 Interactive web application for comparing the experimental BESS scheduling results.

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Contents of Attachments

/	
— README.md	run instructions
— requirements.txt	list of required Python packages
— BESS_schedule.ipynb	main Jupyter notebook
— BESS_schedule.html	notebook with visible outputs
— data	directory with saved data and computed results
— cache_data.pkl	cached input market data
— forecast_extra_update_features.pkl	forecasting models
— forecast_extra_update_features_with_mapie.pkl	MAPIE-based models
— scenarios.pkl	saved scenarios
— src	source code
— data_loader.py	loading data
— evaluation.py	functions for evaluating forecasts and schedules
— feature_maker.py	feature engineering for forecasting models
— naive_schedule.py	implementation of simple baseline schedules
— plot_results.py	functions for plotting results
— price_forecast.py	implementation of forecasting models
— scenarios.py	scenario generation for robust scheduling
— solver.py	optimization models for BESS schedule creation
— thesis.pdf	thesis text in pdf